School Worksheet

Class 12 Mathematics, Chapter 2: Inverse Trigonometric Functions

Maximum marks: **43** **20** questions

Name _____ Date _____ Score _____ / 43

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

The principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$ is:

(A) $-\frac{\pi}{6}$ **(B)** $\frac{7\pi}{6}$

(C) $\frac{5\pi}{6}$ **(D)** $\frac{\pi}{6}$

2 MCQ — 1 mark

The value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$ is:

(A) $\frac{7\pi}{6}$ **(B)** $\frac{5\pi}{6}$

(C) $\frac{\pi}{6}$ **(D)** $-\frac{\pi}{6}$

3 MCQ — 1 mark

The domain of the function $y = \sin^{-1}(2x)$ is:

(A) $[-1, 1]$ **(B)** $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(C) $[-2, 2]$ **(D)** $[0, 1]$

4 MCQ — 1 mark

The range of the principal value branch of $\sec^{-1} x$ is:

(A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ **(B)** $(0, \pi)$

(C) $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ **(D)** $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

5 MCQ — 1 mark

The principal value of $\sec^{-1}\left(\sqrt{2}\right) + 2\,\mathrm{cosec}^{-1}\left(-\sqrt{2}\right)$ is:

(A) $-\frac{\pi}{2}$ **(B)** $-\frac{\pi}{4}$

(C) $\frac{\pi}{4}$ **(D)** $\frac{\pi}{2}$

Q No.	Question	Marks
6	Assertion-Reason **Assertion (A):** The domain of $y = \cos^{-1} x$ is $[-1, 1]$. **Reason (R):** The range of the principal value branch of $y = \cos^{-1} x$ is $[0, \pi] - \left\{\frac{\pi}{2}\right\}$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Find the value of $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$.	2 marks

Level 2 — 16 marks

8	Short answer Evaluate: $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right) + \tan^{-1}\left(\tan\frac{2\pi}{3}\right)$.	2 marks
9	Short answer Find the domain of $f(x) = \sin^{-1}\left(x^2 - 4\right)$.	2 marks
10	Short answer Find the domain of $f(x) = \sin^{-1} x + \cos^{-1}(x - 1)$.	2 marks
11	Short answer Find the value of $\cos^{-1}\left(\cos\frac{13\pi}{6}\right) + \tan^{-1}\left(\tan\frac{7\pi}{6}\right)$.	2 marks
12	Short answer Evaluate: $\sin\left[\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right]$.	2 marks
13	Short answer Find the value of $\tan^{-1}\left[2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right]$.	3 marks
14	Short answer Evaluate: $\sec^2\left(\tan^{-1} 2\right) + \csc^2\left(\cot^{-1} 3\right)$.	3 marks

Level 3 — 19 marks

15	Long answer Draw the graph of $y = \cos^{-1} x$ for its principal value branch. Using the graph, write its domain and range, and state whether the function is increasing or decreasing.	3 marks
16	Long answer Find the value of $\sin^{-1}\left(\sin\frac{4\pi}{5}\right) + \cos^{-1}\left(\cos\frac{4\pi}{3}\right) + \tan^{-1}\left(\tan\frac{3\pi}{4}\right)$.	3 marks
17	Long answer Find the domain and range of $f(x) = 2\sin^{-1}(1 - x)$.	3 marks

Q No.	Question	Marks
18	Long answer Write $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ in the simplest form.	3 marks
19	Long answer Write $\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}}$, $0 < x < \pi$, in the simplest form.	3 marks
20	Case study **Case Study:** A lighthouse is 20 m tall. A boat is at a horizontal distance of x m from the foot of the lighthouse. The angle of elevation θ of the top of the lighthouse from the boat is given by $\theta = \tan^{-1}\left(\frac{20}{x}\right)$. **(i)** Find θ when $x = 20\sqrt{3}$ m. — 1 mark **(ii)** Write the domain and range of the principal value branch of $\tan^{-1} x$. — 1 mark **(iii)** Find the distance x of the boat when $\theta = \frac{\pi}{3}$. Also find the principal value of $\tan^{-1}\left(-\sqrt{3}\right)$. — 2 marks	4 marks

Answer key

Chapter 2: Inverse Trigonometric Functions, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) $-\frac{\pi}{6}$
2	(B) $\frac{5\pi}{6}$
3	(B) $\left[-\frac{1}{2}, \frac{1}{2}\right]$
4	(C) $[0, \pi] - \left\{\frac{\pi}{2}\right\}$
5	(B) $-\frac{\pi}{4}$
6	(C) A is true, but R is false
7	$\frac{2\pi}{5}$

Level 2

8	$-\frac{5\pi}{12}$
9	$\left[-\sqrt{5}, -\sqrt{3}\right] \cup \left[\sqrt{3}, \sqrt{5}\right]$
10	$[0, 1]$
11	$\frac{\pi}{3}$
12	1
13	$\frac{\pi}{4}$
14	15

Level 3

15	Domain $[-1, 1]$; range $[0, \pi]$; $y = \cos^{-1} x$ is a decreasing function.
16	$\frac{37\pi}{60}$
17	Domain $[0, 2]$; range $[-\pi, \pi]$
18	$\tan^{-1} x$
19	$\frac{x}{2}$
20	(i) $\theta = \frac{\pi}{6}$ (ii) Domain $\mathbb{R}$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (iii) $x = \frac{20\sqrt{3}}{3}$ m ≈ 11.55 m; $\tan^{-1}\left(-\sqrt{3}\right) = -\frac{\pi}{3}$